

Laboratoire 4 : Surveillance de réseau avec Wireshark et Zenmap

Objectifs pédagogiques :

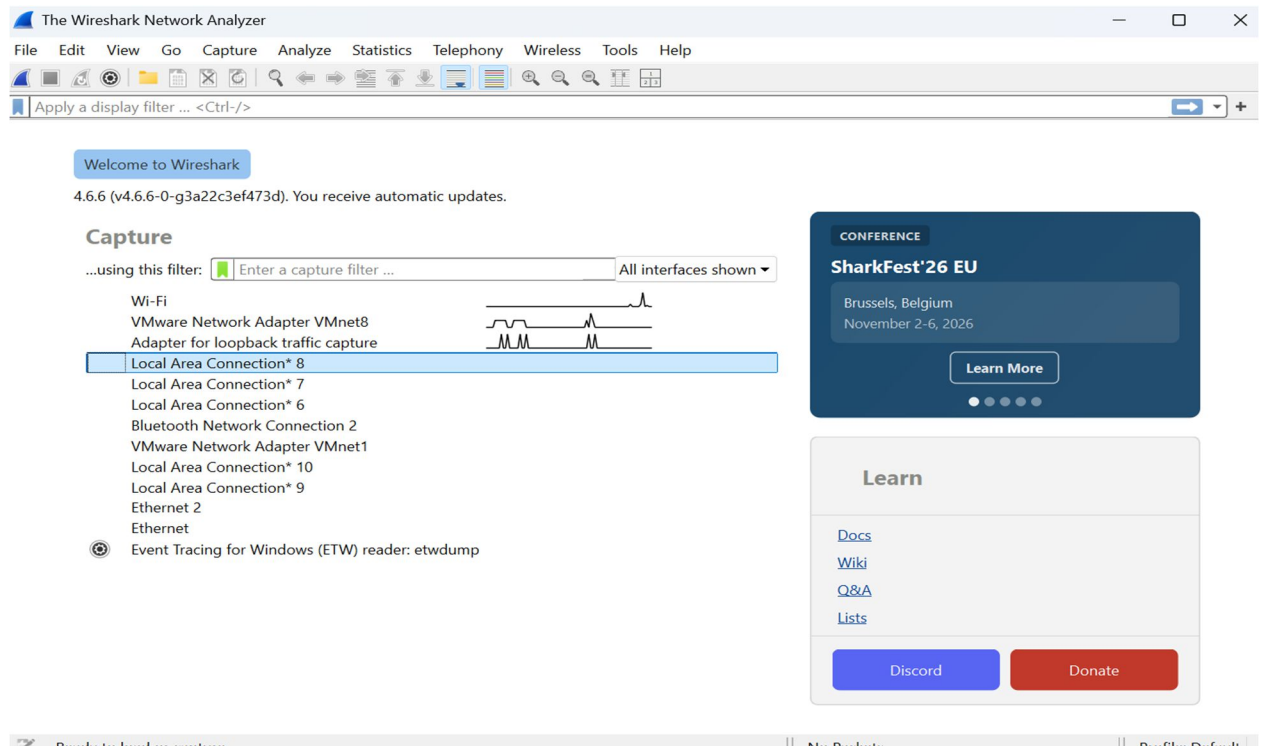
Capturer et analyser du trafic réseau avec Wireshark. Réaliser des scans réseau avec Zenmap (interface de Nmap). Identifier des ports ouverts, services actifs, et anomalies potentielles. Comprendre les enjeux de la surveillance réseau en cybersécurité.

Partie 1 : Analyse de trafic réseau avec Wireshark

Étape 1 : Lancement de la capture

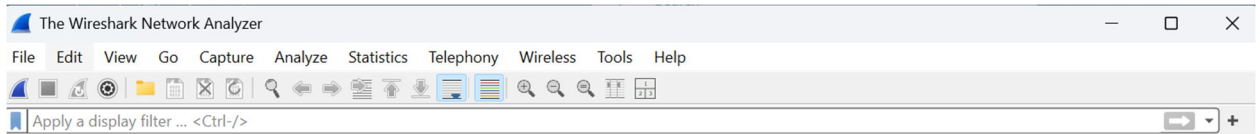
a) Ouvrir Wireshark.

Nous avons ici interface de wireshark



b) 2- Sélectionner l'interface réseau active (ici wifi).

Nous choisissons ici notre carte réseau wifi pour faire des captures. Par la suite nous allons cliquer sur start pour commencer les captures



Welcome to Wireshark

4.6.6 (v4.6.6-0-g3a22c3ef473d). You receive automatic updates.

Capture

...using this filter:

All interfaces shown

Wi-Fi	
VMware Network Adapter VMnet8	
VMware Network Adapter VMnet1	
Ethernet 2	
Local Area Connection* 8	
Local Area Connection* 7	



Étape 2 : Générer du trafic

Il est question pour nous dans cette étape de générer du trafic puisque les capture wireshark sont déjà en cour nous allons ouvrir un google chrome et accéder a <https://www.google.com> et effectuer une recherche

No.	Time	Source	Destination	Protocol	Length	Info
7984	42.736598200	2001:4958:4021:7a01...	fe80::9a42:65ff:fef...	ICMPv6	86	Neighbor Advertisement 2001:4958:4021:7a01:5c73:65...
7985	42.910563600	2001:4958:4021:7a01...	2620:1ec:33::11	TLSv1.2	782	Application Data
7986	42.910794100	2001:4958:4021:7a01...	2620:1ec:33::11	TLSv1.2	1044	Application Data
7987	42.917313900	13.89.178.26	192.168.2.13	TCP	56	443 → 59118 [ACK] Seq=11108 Ack=169180 Win=4194560
7988	42.917396000	192.168.2.13	13.89.178.26	TCP	54	[TCP Dup ACK 1405#1] 59118 → 443 [ACK] Seq=169180
7989	42.930065400	2620:1ec:33::11	2001:4958:4021:7a01...	TCP	78	443 → 51201 [ACK] Seq=13408 Ack=33413 Win=16381 Len=
7990	42.930068600	2620:1ec:33::11	2001:4958:4021:7a01...	TCP	78	443 → 51201 [ACK] Seq=13408 Ack=34383 Win=16384 Len=
7991	42.952424300	13.89.178.26	192.168.2.13	TCP	56	[TCP Dup ACK 1402#1] 443 → 59118 [ACK] Seq=11601 A
7992	42.964138100	2620:1ec:33::11	2001:4958:4021:7a01...	TLSv1.2	1135	Application Data
7993	43.009624700	2001:4958:4021:7a01...	2620:1ec:33::11	TCP	74	51201 → 443 [ACK] Seq=34383 Ack=14465 Win=251 Len=
7994	43.484012000	2001:4958:4021:7a01...	2001:4958:732::1	DNS	94	Standard query 0xb8a0 HTTPS www.polymtl.ca
7995	43.484285900	2001:4958:4021:7a01...	2001:4958:732::1	DNS	94	Standard query 0x58d6 AAAA www.polymtl.ca
7996	43.484474800	2001:4958:4021:7a01...	2001:4958:732::1	DNS	94	Standard query 0x03a4 A www.polymtl.ca
7997	43.544159000	2001:4958:4021:7a01...	2001:4958:733::1	DNS	94	Standard query 0x5992 HTTPS www.polymtl.ca

> Frame 7103: Packet, 466 bytes on wire (3728 bits), 466 bytes captured (3728 bits) on interface 0

> Ethernet II, Src: Intel_56:6c:ba (a8:7e:ea:56:6c:ba), Dst: Sagem

> Internet Protocol Version 4, Src: 192.168.2.13, Dst: 204.79.197.1

> Transmission Control Protocol, Src Port: 57493, Dst Port: 80, Seq=

> Hypertext Transfer Protocol

0000 98 42 65 fd b4 68 a8 7e ea 56 6c ba 08 00 45 00 ·Be·h·

0010 01 c4 d6 b3 40 00 80 06 00 00 c0 a8 02 0d cc 4f ····@·

0020 c5 cb e0 95 00 50 43 06 fb 39 e0 14 a1 0b 50 18 ····PC

0030 00 ff 56 87 00 00 47 45 54 20 2f 6f 63 73 70 2f ···V··G

0040 4d 46 51 77 55 6a 42 51 4d 45 34 77 54 44 41 4a MFQwUjB

0050 42 67 55 72 44 67 4d 43 47 67 55 41 42 42 52 30 BgUrDgM

0060 54 42 45 56 59 6b 6c 58 37 41 39 79 4c 6f 4c 44 TBEVYkL

0070 39 68 71 6d 43 57 44 78 46 67 51 55 33 70 47 47 9hqmCWD

0080 53 4c 65 68 4d 56 6b 78 38 55 74 66 42 36 6e 63 SLehMVk

0090 69 48 6e 61 71 48 59 43 45 7a 4d 41 41 41 50 50 iHnagHY

00a0 4d 79 42 6c 4e 25 32 42 35 43 72 6b 38 41 41 41 MyB1N%2

00b0 41 41 41 41 38 25 33 44 20 48 54 54 50 2f 31 2e AAAA8%3

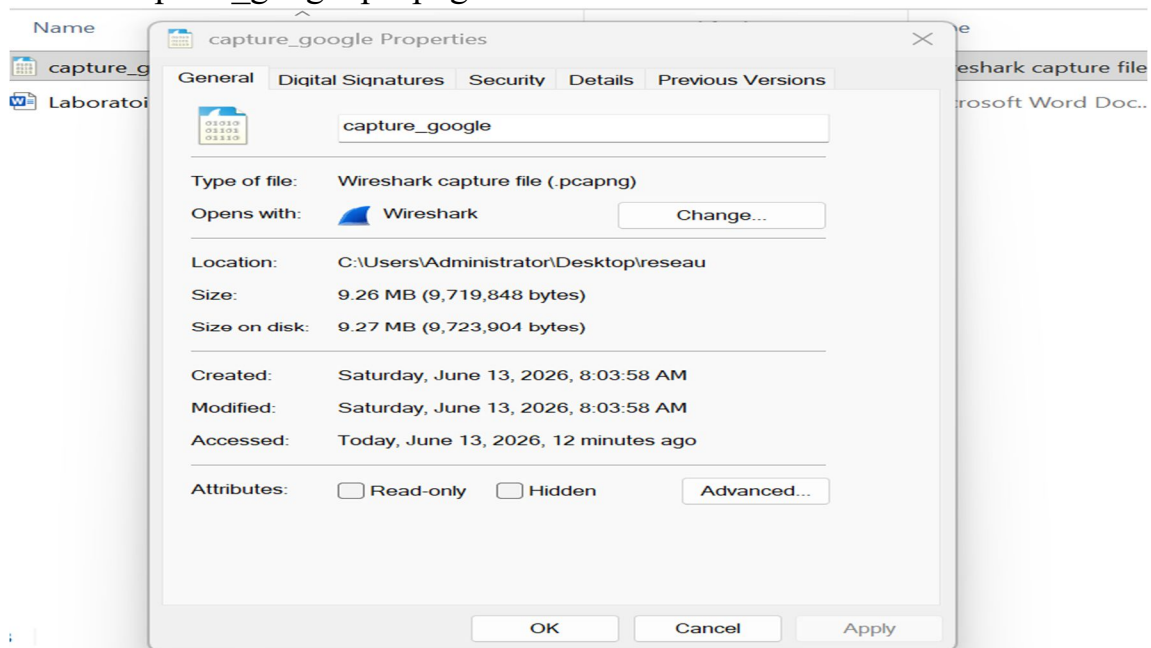
00c0 31 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 1··Cach

00d0 3a 20 6d 61 78 2d 61 67 65 20 3d 20 38 36 34 30 : max-a

00e0 30 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 4b 0··Conn

Étape 3 : Arrêt et sauvegarde

dans cette étape nous devons arrêter la capture et sauvegarder le fichier sous le nom capture_google.pcapng



Étape 4 : Application de filtres

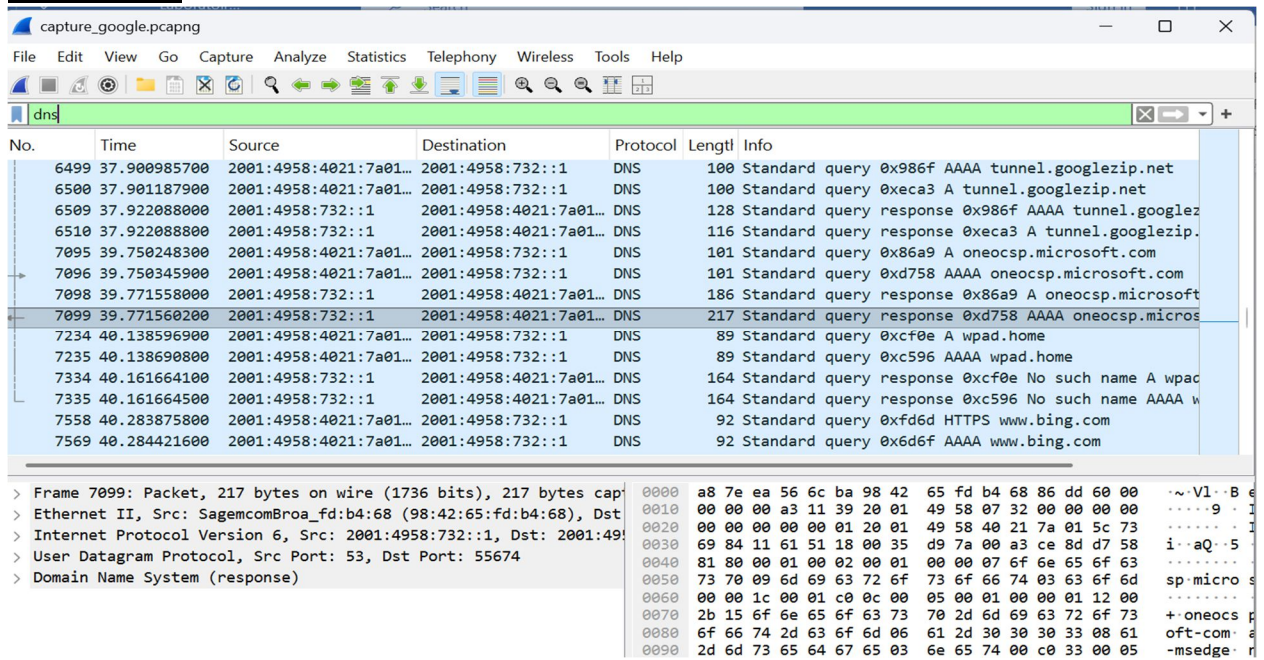
Dans cette partie nous devons appliquer des filtres pour avoir des information spécifique sur le protocole http, dns, tcp, et l'adresse ip 192.168.2.13

Filtre http

No.	Time	Source	Destination	Protocol	Length	Info
7103	39.796323900	192.168.2.13	204.79.197.203	HTTP	466	GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGGUABBR0TBEVYk1X7A9yl
7106	39.818896300	204.79.197.203	192.168.2.13	OCSP	1369	Response
7108	39.825678500	192.168.2.13	204.79.197.203	HTTP	466	GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGGUABBR0TBEVYk1X7A9yl
7111	39.848543100	204.79.197.203	192.168.2.13	OCSP	1369	Response
7113	39.856325600	192.168.2.13	204.79.197.203	HTTP	464	GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGGUABBR0TBEVYk1X7A9yl
7116	39.878715700	204.79.197.203	192.168.2.13	OCSP	1369	Response
7118	39.887835500	192.168.2.13	204.79.197.203	HTTP	464	GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGGUABBR0TBEVYk1X7A9yl
7122	39.911562500	204.79.197.203	192.168.2.13	OCSP	1369	Response

> Frame 7103: Packet, 466 bytes on wire (3728 bits), 466 bytes captured on interface 0	
Ethernet II, Src: Intel_56:6c:ba (a8:7e:ea:56:6c:ba), Dst: Sage	0000 98 42 65 fd b4 68 a8 7e ea 56 6c ba 08 00 45 00
Internet Protocol Version 4, Src: 192.168.2.13, Dst: 204.79.197.203	0010 01 c4 d6 b3 40 00 80 06 00 00 c0 a8 02 0d cc 4f
Transmission Control Protocol, Src Port: 57493, Dst Port: 80, Seq: 300000000, Win: 65535, Len: 0	0020 c5 cb e0 95 00 50 43 06 fb 39 e0 14 a1 0b 50 18
Hypertext Transfer Protocol	0030 00 ff 56 87 00 00 47 45 54 20 2f 6f 63 73 70 2f
	0040 4d 46 51 77 55 6a 42 51 4d 45 34 77 54 44 41 4a
	0050 42 67 55 72 44 67 4d 43 47 67 55 41 42 42 52 30

Filtre DNS

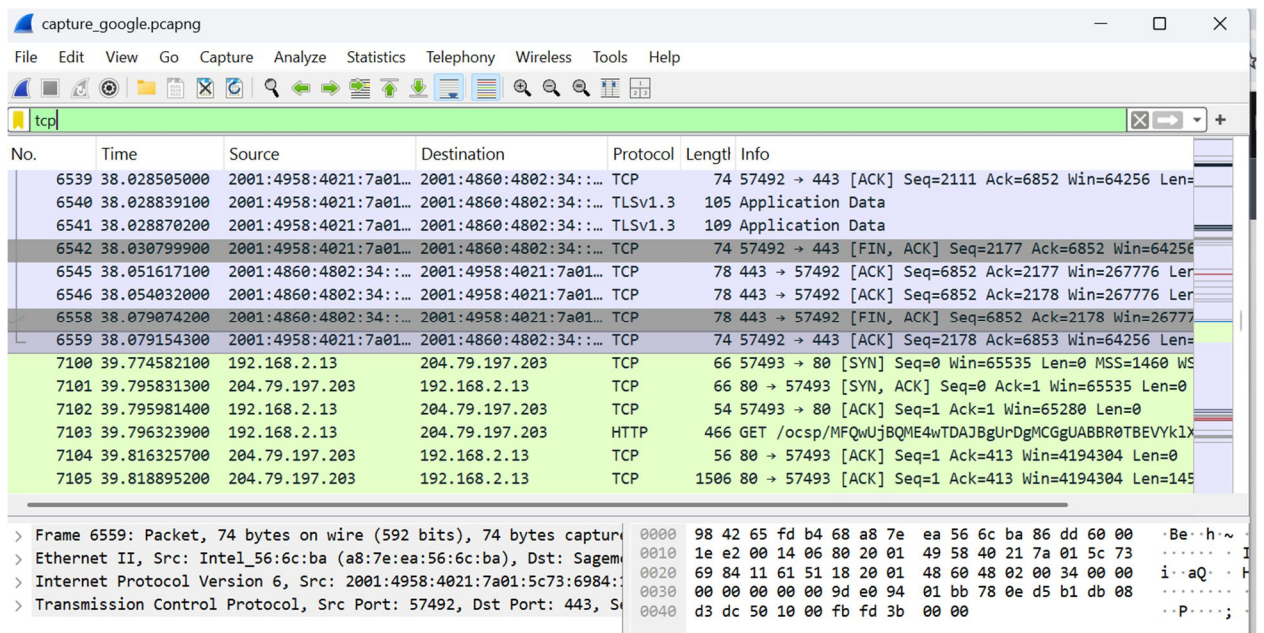


The screenshot shows a Wireshark capture of network traffic filtered by the 'dns' protocol. The packet list on the left shows several DNS queries and responses. The selected packet (No. 7099) is a DNS response from 2001:4958:4021:7a01::1 to 2001:4958:732::1. The packet details pane on the right shows the structure of the DNS response, including the query type (AAAA), the response code (No such name), and the domain name (www.bing.com). The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
6499	37.900985700	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	100	Standard query 0x986f AAAA tunnel.googlezip.net
6500	37.901187900	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	100	Standard query 0xeca3 A tunnel.googlezip.net
6509	37.922088000	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	128	Standard query response 0x986f AAAA tunnel.googlezip.net
6510	37.922088800	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	116	Standard query response 0xeca3 A tunnel.googlezip.net
7095	39.750248300	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	101	Standard query 0x86a9 A oneocsp.microsoft.com
7096	39.750345900	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	101	Standard query 0xd758 AAAA oneocsp.microsoft.com
7098	39.771558000	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	186	Standard query response 0x86a9 A oneocsp.microsoft.com
7099	39.771560200	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	217	Standard query response 0xd758 AAAA oneocsp.microsoft.com
7234	40.138596900	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	89	Standard query 0xcfc0e A wpad.home
7235	40.138690800	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	89	Standard query 0xc596 AAAA wpad.home
7334	40.161664100	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	164	Standard query response 0xcfc0e No such name A wpad.home
7335	40.161664500	2001:4958:732::1	2001:4958:4021:7a01::1	DNS	164	Standard query response 0xc596 No such name AAAA wpad.home
7558	40.283875800	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	92	Standard query 0xfd6d HTTPS www.bing.com
7569	40.284421600	2001:4958:4021:7a01::1	2001:4958:732::1	DNS	92	Standard query 0xd6f AAAA www.bing.com

> Frame 7099: Packet, 217 bytes on wire (1736 bits), 217 bytes captured on interface 0
> Ethernet II, Src: SagemcomBroadband_8d:6b:b3, Dst: 2001:4958:4021:7a01::1
> Internet Protocol Version 6, Src: 2001:4958:732::1, Dst: 2001:4958:4021:7a01::1
> User Datagram Protocol, Src Port: 53, Dst Port: 55674
> Domain Name System (response)

Filtre 192.168.2.13



The screenshot shows a Wireshark capture of network traffic filtered by the 'tcp' protocol. The packet list on the left shows several TCP connections. The selected packet (No. 6559) is a TCP ACK from 2001:4958:4021:7a01::1 to 2001:4860:4802:34::1. The packet details pane on the right shows the structure of the TCP segment, including the sequence number (Seq=2178), acknowledgment number (Ack=6853), and window size (Win=64256). The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
6539	38.028505000	2001:4958:4021:7a01::1	2001:4860:4802:34::1	TCP	74	57492 → 443 [ACK] Seq=2111 Ack=6852 Win=64256 Len=0
6540	38.028839100	2001:4958:4021:7a01::1	2001:4860:4802:34::1	TLSv1.3	105	Application Data
6541	38.028870200	2001:4958:4021:7a01::1	2001:4860:4802:34::1	TLSv1.3	109	Application Data
6542	38.030799900	2001:4958:4021:7a01::1	2001:4860:4802:34::1	TCP	74	57492 → 443 [FIN, ACK] Seq=2177 Ack=6852 Win=64256 Len=0
6545	38.051617100	2001:4860:4802:34::1	2001:4958:4021:7a01::1	TCP	78	443 → 57492 [ACK] Seq=6852 Ack=2177 Win=267776 Len=0
6546	38.054032000	2001:4860:4802:34::1	2001:4958:4021:7a01::1	TCP	78	443 → 57492 [ACK] Seq=6852 Ack=2178 Win=267776 Len=0
6558	38.079074200	2001:4860:4802:34::1	2001:4958:4021:7a01::1	TCP	78	443 → 57492 [FIN, ACK] Seq=6852 Ack=2178 Win=267776 Len=0
6559	38.079154300	2001:4958:4021:7a01::1	2001:4860:4802:34::1	TCP	74	57492 → 443 [ACK] Seq=2178 Ack=6853 Win=64256 Len=0
7100	39.774582100	192.168.2.13	204.79.197.203	TCP	66	57493 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=0
7101	39.795831300	204.79.197.203	192.168.2.13	TCP	66	80 → 57493 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
7102	39.795981400	192.168.2.13	204.79.197.203	TCP	54	57493 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
7103	39.796323900	192.168.2.13	204.79.197.203	HTTP	466	GET /ocsp/MFQwUjBQME4wTDAJBgUrDgMCGGUABBR0TBEVYk1X...
7104	39.816325700	204.79.197.203	192.168.2.13	TCP	56	80 → 57493 [ACK] Seq=1 Ack=413 Win=4194304 Len=0
7105	39.818895200	204.79.197.203	192.168.2.13	TCP	1506	80 → 57493 [ACK] Seq=1 Ack=413 Win=4194304 Len=145

> Frame 6559: Packet, 74 bytes on wire (592 bits), 74 bytes captured on interface 0
> Ethernet II, Src: Intel_56:6c:ba (a8:7e:ea:56:6c:ba), Dst: SagemcomBroadband_8d:6b:b3
> Internet Protocol Version 6, Src: 2001:4958:4021:7a01:5c73:6984::1, Dst: 2001:4860:4802:34::1
> Transmission Control Protocol, Src Port: 57492, Dst Port: 443, Seq: 2178, Ack: 6853, Win: 64256, Len: 0

Repondons aux questions

- Quelle est l'adresse ip de google : cette adresse est 142.250.137.157 pour la trouver nous avons utiliser le filtre ci-dessous

capture_google.pcapng						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
ip.addr == 142.250.0.0/15						
No.	Time	Source	Destination	Protocol	Length	Info
5628	37.720375000	192.168.2.13	142.250.137.157	QUIC	1292	Initial, DCID=c075b6107cbe16a7, PKN: 1, PADDING, C
5629	37.720519300	192.168.2.13	142.250.137.157	QUIC	1292	Initial, DCID=c075b6107cbe16a7, PKN: 2, PADDING, F
5746	37.753398500	142.250.137.157	192.168.2.13	QUIC	82	Initial, SCID=e075b6107cbe16a7, PKN: 1, ACK
5747	37.753420400	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 2, ACK, PADDI
5748	37.753422700	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 3, CRYPTO, PA
5749	37.753427000	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 4, CRYPTO, PA
5750	37.754245200	192.168.2.13	142.250.137.157	QUIC	1292	Initial, DCID=e075b6107cbe16a7, PKN: 3, ACK, PADDI
5909	37.784172300	142.250.137.157	192.168.2.13	QUIC	1259	Protected Payload (KP0)
5968	37.788457500	192.168.2.13	142.250.137.157	QUIC	82	Handshake, DCID=e075b6107cbe16a7
5969	37.788497400	192.168.2.13	142.250.137.157	QUIC	79	Handshake, DCID=e075b6107cbe16a7
6318	37.827434300	142.250.137.157	192.168.2.13	QUIC	153	Protected Payload (KP0)
6337	37.828220500	192.168.2.13	142.250.137.157	QUIC	83	Handshake, DCID=e075b6107cbe16a7
6362	37.830126700	142.250.137.157	192.168.2.13	QUIC	1292	Handshake, SCID=e075b6107cbe16a7
6363	37.830128600	142.250.137.157	192.168.2.13	QUIC	1292	Handshake, SCID=e075b6107cbe16a7

> Frame 5628: Packet, 1292 bytes on wire (10336 bits), 1292 bytes	0000	98 42 65 fd b4 68 a8 7e	ea 56 6c ba 08 00 45 00	Be . h . .
> Ethernet II, Src: Intel_56:6c:ba (a8:7e:ea:56:6c:ba), Dst: Sagem	0010	04 fe 5f a5 40 00 80 11	00 00 c0 a8 02 0d 8e fa	 @ . . .
> Internet Protocol Version 4, Src: 192.168.2.13, Dst: 142.250.137	0020	89 9d f5 0b 01 bb 04 ea	e0 48 c1 00 00 00 01 08	
> User Datagram Protocol, Src Port: 62731, Dst Port: 443	0030	c0 75 b6 10 7c be 16 a7	00 00 44 d0 7d 45 0b dd	. u
> QUIC IETF	0040	61 05 6b 09 76 2a c8 3d	6e bf 5a b3 f2 72 af df	a . k . v * .
	0050	85 0e 35 f3 f9 98 f2 b3	0a 03 35 fc 55 7d 3f f9	. . 5

- Combien de requêtes DNS ont été envoyées : nous avons 10771 requettes dns envoyer pour avoir ce nombre nous avons regarder le numero de la derniere requette du filtre dns

10768	60.972264300	2001:4958:732::1	2001:4958:4021:7a01...	DNS	234	Standard query response 0xd10 A browser.events.da
10769	60.972267500	2001:4958:732::1	2001:4958:4021:7a01...	DNS	289	Standard query response 0x24bd AAAA browser.events
10771	61.001530900	2001:4958:733::1	2001:4958:4021:7a01...	DNS	234	Standard query response 0xdf10 A browser.events.da

- Quelle est le port utiliser pour https : le port utiliser pour https est 443 seulement notre navigateur ici utilise le protocole QUIC pour accéder a la connexion

5745	37.749662600	2001:4958:4021:7a01...	2001:4860:482c:7700...	QUIC	108	Protected Payload (KP0), DCID=eb3e2b40f49d30d9
5746	37.753398500	142.250.137.157	192.168.2.13	QUIC	82	Initial, SCID=e075b6107cbe16a7, PKN: 1, ACK
5747	37.753420400	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 2, ACK, PADDI
5748	37.753422700	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 3, CRYPTO, PA
5749	37.753427000	142.250.137.157	192.168.2.13	QUIC	1292	Initial, SCID=e075b6107cbe16a7, PKN: 4, CRYPTO, PA
5750	37.754245200	192.168.2.13	142.250.137.157	QUIC	1292	Initial, DCID=e075b6107cbe16a7, PKN: 3, ACK, PADDI
5751	37.756808600	2001:4860:482c:7700...	2001:4958:4021:7a01...	QUIC	88	Protected Payload (KP0)
5752	37.756968600	2001:4958:4021:7a01...	2001:4860:482c:7700...	QUIC	100	Protected Payload (KP0), DCID=eb3e2b40f49d30d9
5756	37.759896500	2001:4860:482c:7700...	2001:4958:4021:7a01...	QUIC	210	Protected Payload (KP0)

- Quelle est la différence entre le protocole TCP et le protocole UDP

TCP est fiable : il garantit que les données arrivent dans l'ordre, sans perte (accusés de réception).

UDP est rapide : il envoie les données sans vérification ni garantie de livraison.

- Pourquoi est-il utile de filtrer les paquets ?

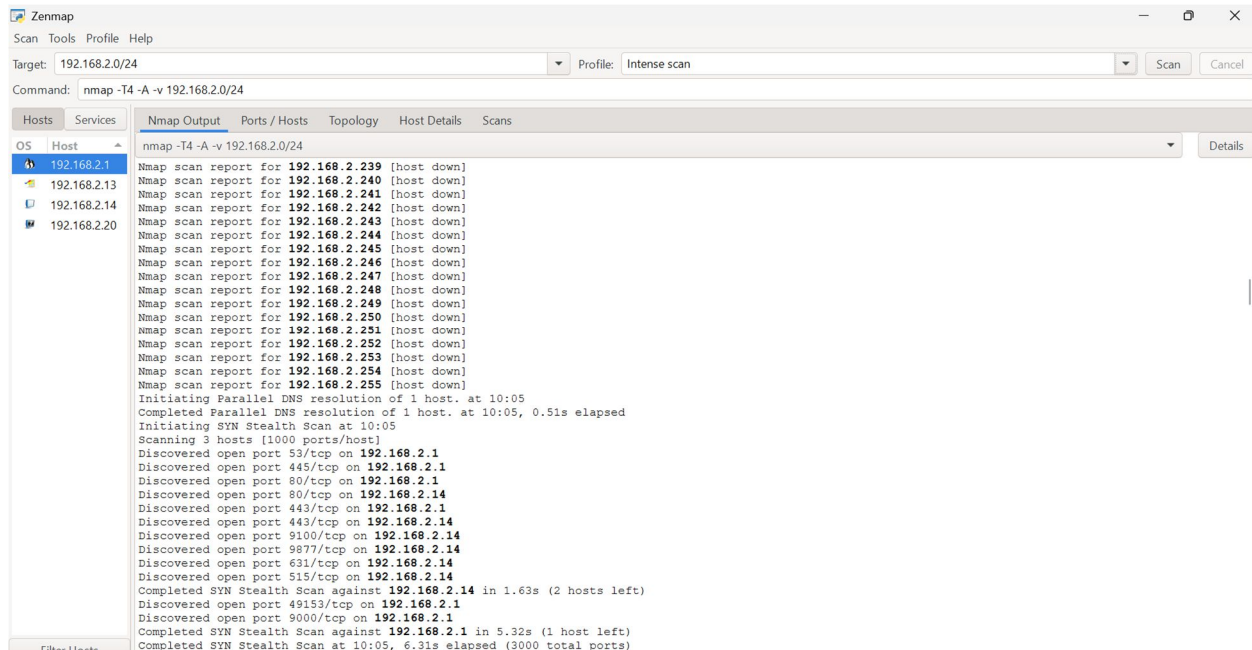
En réseau, des milliers de paquets circulent en permanence : trafic système, publicités, mises à jour, protocoles internes. Filtrer permet de : Isoler uniquement le trafic pertinent, Détecter des anomalies ou intrusions, Analyser un protocole précis , Gagner du temps en évitant de parcourir des milliers de lignes inutiles

Partie 2 : Scan de réseau avec Zenmap (12 pts)

Étape 1 : Scan du réseau local

1. Ouvrir Zenmap.
2. Dans "Target", entrer : 192.168.1.0/24 (adapter selon votre réseau).
3. Dans "Profile", choisir : Intense scan.
4. Lancer le scan.
5. Identifier les hôtes détectés.

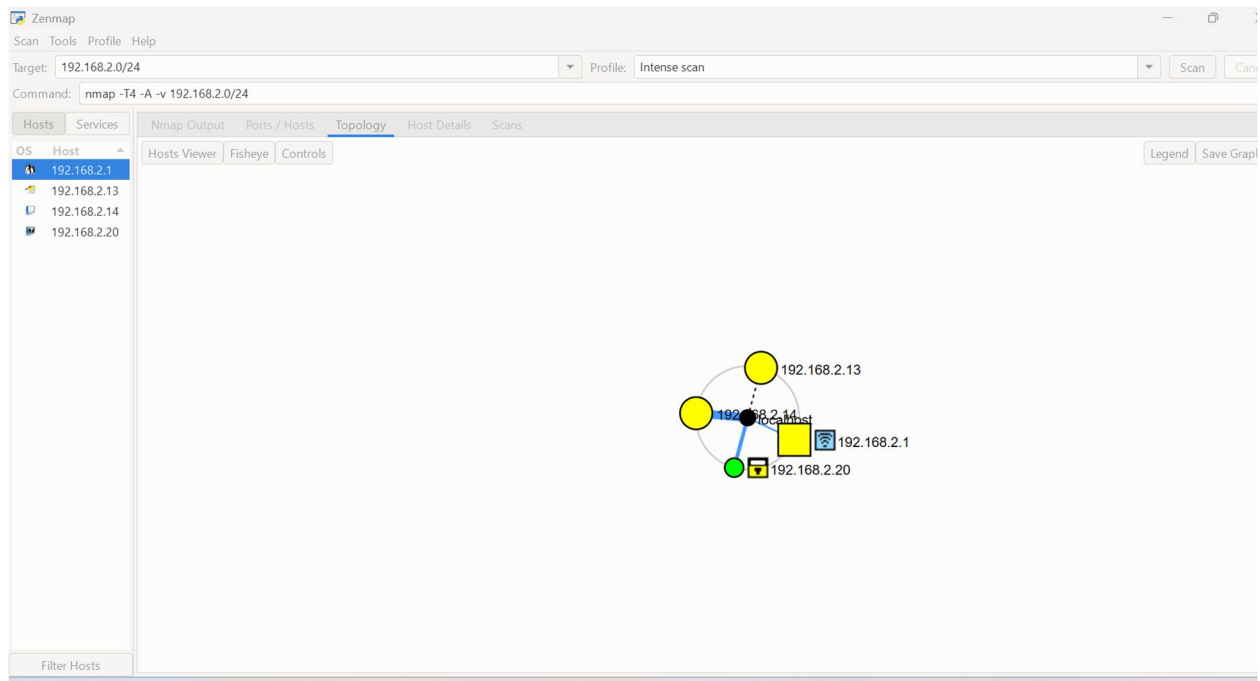
Dans cette partie nous allons scanner le réseau avec Zenmap tout d'abord nous allons ouvrir zenmap et dans Target nous entrons 192.168.2/24 dans profile on choisi intense scan et nous lançons le scan. Et identifions les hôtes détectés



Les hôtes détectés sont 192.168.2.1, 192.168.2.13, 192.168.2.14, 192.168.2.20

Étape 2 : Affichage de la topologie

Nous allons analyser la topologie de notre réseau scanner



Le réseau 192.168.2.0/24 présente une architecture en étoile centrée sur le routeur 192.168.2.1 (carré jaune), qui constitue le point d'accès central. Trois équipements y sont connectés :

- 192.168.2.13 (mon PC Windows 11) cercle jaune, lien en pointillés
- 192.168.2.14 (imprimante Canon) cercle jaune
- 192.168.2.20

Étape 3 : Scan ciblé d'une machine (comme par exemple Metasploitable 2)

1. Choisir une machine détectée (ex. 192.168.1.5).

2. Lancer le scan suivant :

o `nmap -sS -sV 192.168.1.5`

o Puis : `nmap --script vuln 192.168.1.5`

ici nous allons scanner notre machine imprimante 192.168.2.14 avec ces deux commande `nmap -sS -sV 192.168.2.14` et `nmap --script vuln 192.168.2.14`

`nmap -sS -sV 192.168.2.14`

Zenmap

Scan Tools Profile Help

Target: 192.168.2.14

Command: nmap -sS -sV 192.168.2.14

Hosts Services

Nmap Output

Ports / Hosts Topology Host Details Scans

OS Host

192.168.2.1

192.168.2.13

192.168.2.14

192.168.2.20

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-13 12:49 -0400

```

SF: index.html" > \r\n\x20<head>\r\n<html>\r\n") &r(HTTPOptions,5A,"H
SF: TTP/1.1\x20405\x20Method\x20Not\x20Allowed\r\nMIME-Version:\x201.0\r\n
SF: nAllow:\x20GET,HEAD\r\nConnection:\x20Close\r\n\r\n") &r(FourOhFourReque
SF: st,160,"HTTP/1.1\x20404\x20Not\x20Found\r\nMIME-Version:\x201.0\r\nCo
SF: nent-Type:\x20text/html\r\nConnection:\x20Close\r\n\r\n<HTML>\r\n\x20
SF: i\x20<HEAD>\r\n\x20\x20\x20\x20-META\x20http-equiv=\x20Content-Type\x20
SF: NTEXT=\x20text/html;\x20charset=EUC-JP" > \r\n\x20\x20\x20\x20<TITLE>HTT
SF: P/1.1\x20404\x20Not\x20Found</TITLE>\r\n\x20\x20<HEAD>\r\n\r\n\x20\x20<
SF: BODY>\r\n\x20\x20\x20\x20<H1>404\x20Not\x20Found</H1><BR>\r\n\x20\x20\x20
SF: 20\x20<HR>\r\n\x20\x20\x20\x20\x20A\x20specification\x20file\x20is\x20not\x20
SF: 20found<DR>\r\n\x20\x20<BODY>\r\n<HTML>\r\n") &r(GenericLines,171,"HTT
SF: P/1.1\x20400\x20Bad\x20Request\r\nMIME-Version:\x201.0\r\nContent-Type
SF: =\x20text/html\r\nConnection:\x20Close\r\n\r\n<HTML>\r\n\x20\x20<HEAD>
SF: \r\n\x20\x20\x20\x20-META\x20http-equiv=\x20Content-Type\x20CONTENT=\x20
SF: ext/html;\x20charset=EUC-JP" > \r\n\x20\x20\x20\x20<TITLE>HTTP/1.1\x204
SF: 00\x20Bad\x20Request</TITLE>\r\n\x20\x20<HEAD>\r\n\r\n\x20\x20<BODY>\r
SF: \r\n\x20\x20\x20\x20<H1>400\x20Bad\x20Request</H1><BR>\r\n\x20\x20\x20\x20
SF: 0<HR>\r\n\x20\x20\x20\x20An\x20error\x20is\x20in\x20description\x20of\x20a
SF: 20a\x20request\x20line<BR>\r\n\x20\x20<BODY>\r\n<HTML>\r\n") &r(RTSPre
SF: quest,192,"HTTP/1.1\x20505\x20HTTP\x20Version\x20Not\x20Supported\r\nM
SF: MIME-Version:\x201.0\r\nContent-Type:\x20text/html\r\nConnection:\x20Cl
SF: ose\r\n\r\n<HTML>\r\n\x20\x20<HEAD>\r\n\x20\x20\x20\x20-META\x20http-eg
SF: uiv=\x20Content-Type\x20CONTENT=\x20text/html;\x20charset=EUC-JP" > \r\n\x20
SF: 20\x20\x20\x20<TITLE>HTTP/1.1\x20505\x20HTTP\x20Version\x20Not\x20Supp
SF: orted</TITLE>\r\n\x20\x20<HEAD>\r\n\r\n\x20\x20<BODY>\r\n\x20\x20\x20\x20
SF: 20<H1>505\x20HTTP\x20Version\x20Not\x20Supported\r\n\x20\x20\x20\x20<H
SF: 20>\r\n\x20\x20\x20\x20requested\x20HTTP\x20version\x20cannot\x20
SF: be\x20used<BR>\r\n\x20\x20<BODY>\r\n<HTML>\r\n");
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF: Port 631-TCP:V=7.99&T=SSL&I=7&D=6/13&Time=6A2D8A19&P=i686-pc-windows-win
SF: dows&r(GenericLines,171,"HTTP/1.1\x20400\x20Bad\x20Request\r\nMIME-Ver
SF: sion:\x201.0\r\nContent-Type:\x20text/html\r\nConnection:\x20Close\r\n
SF: i\r\n<HTML>\r\n\x20\x20<HEAD>\r\n\r\n\x20\x20\x20\x20-META\x20http-equiv=\x20
SF: Content-Type\x20CONTENT=\x20text/html;\x20charset=EUC-JP" > \r\n\x20\x20

```

```

SF: \r\n<HTML>\r\n\x20\x20<HEAD>\r\n\x20\x20\x20\x20-META\x20http-equiv=\x20C
SF: ontent-Type\x20CONTENT=\x20text/html;\x20charset=EUC-JP" > \r\n\x20\x20
SF: x20\x20<TITLE>HTTP/1.1\x20400\x20Bad\x20Request</TITLE>\r\n\x20\x20<H
SF: EAD>\r\n\r\n\x20\x20<BODY>\r\n\x20\x20\x20\x20<H1>400\x20Bad\x20Request
SF: </H1><BR>\r\n\x20\x20\x20\x20<HR>\r\n\x20\x20\x20\x20An\x20error\x20is\
SF: x20in\x20description\x20of\x20a\x20request\x20line<BR>\r\n\x20\x20</BOD
SF: Y>\r\n\r\n<HTML>\r\n") &r(GetRequest,1DF,"HTTP/1.1\x20200\x20OK\r\nMIME-Ver
SF: sion:\x201.0\r\nLast-Modified:\x20Mon,\x2015\x20Jan\x202007\x2007:45:
SF: 41\x20GMT\r\nAccept-Ranges:\x20bytes\r\nContent-Length:\x20277\r\nETag:
SF: \x20"0000ee2b-00000115"\r\nKeep-Alive:\x20timeout=30\r\nConnection:\x20
SF: 20Keep-Alive\r\n\r\n\r\n<html>\r\n\x20\x20<head>\r\n\x20\x20\x20\x20<meta\x20
SF: 20http-equiv=\x20Content-Type\x20content=\x20text/html;\x20charset=Shift_
SF: JIS" > \r\n\x20\x20\x20\x20<meta\x20http-equiv=author\x20content=\x20Can
SF: on" > \r\n\r\n\r\n\x20\x20\x20\x20<meta\x20http-equiv=\x20Pragma\x20conte
SF: nt=\x20no-cache" > \r\n\r\n\x20\x20\x20\x20<meta\x20http-equiv=\x20refresh" > \r\n\r\n
SF: content=\x200;url=index.html" > \r\n\x20\x20</head>\r\n</html>\r\n") &r(HT
SF: TPOptions,59,"HTTP/1.1\x20200\x20OK\r\nAllow:\x20OPTIONS,\x20GET,\x20P
SF: OST\r\nConnection:\x20Keep-Alive\r\nContent-Length:\x200\r\n\r\n") &r(RT
SF: SPRequest,192,"HTTP/1.1\x20505\x20HTTP\x20Version\x20Not\x20Supported\
SF: \r\nMIME-Version:\x201.0\r\nContent-Type:\x20text/html\r\nConnection:\x20
SF: 20Close\r\n\r\n\r\n<HTML>\r\n\x20\x20<HEAD>\r\n\x20\x20\x20\x20-META\x20htt
SF: p-equiv=\x20Content-Type\x20CONTENT=\x20text/html;\x20charset=EUC-JP" > \r
SF: \r\n\x20\x20\x20\x20<TITLE>HTTP/1.1\x20505\x20HTTP\x20Version\x20Not\x20
SF: Supported</TITLE>\r\n\x20\x20<HEAD>\r\n\r\n\r\n\x20\x20<BODY>\r\n\x20\x20\x20\
SF: x20\x20<H1>505\x20HTTP\x20Version\x20Not\x20Supported\r\n\x20\x20\x20\x20\
SF: 20<HR>\r\n\r\n\x20\x20\x20\x20The\x20requested\x20HTTP\x20version\x20cannot
SF: be\x20used<BR>\r\n\r\n\x20\x20<BODY>\r\n\r\n<HTML>\r\n");
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF: Port 9877-TCP:V=7.99&I=7&D=6/13&Time=6A2D8A13&P=i686-pc-windows-windows%
SF: r(GenericLines,10,"UNKNOWN\x20COMMAND\r\n") &r(VersionRequest,10,"UNKNOWN\
SF: x20COMMAND\r\n");
MAC Address: 6C:F2:D8:04:7D:D5 (Canon)

```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 158.33 seconds

le scan ciblé de mon imprimante Canon PIXMA, confirmé par l'adresse MAC 6C:F2:D8:04:7D:D5 appartenant à Canon.

Les ports détectés sont les mêmes qu'au scan général :

Port	Service	Description
80/tcp	HTTP	Interface web d'administration
443/tcp	HTTPS	Interface web sécurisée
515/tcp	LPD/Printer	Protocole d'impression classique
631/tcp	IPP (SSL)	Impression via Internet (standard moderne)
9100/tcp	JetDirect	Impression directe en réseau
9877/tcp	Inconnu	Répond "UNKNOWN COMMAND" — service propriétaire Canon

nmap --script vuln 192.168.2.14

The screenshot shows the Zenmap application window. The 'Hosts' tab is active, displaying a list of hosts. The 'Nmap Output' tab is selected, showing the results of the command `nmap --script vuln 192.168.2.14`. The output indicates that the host is up and that 994 closed TCP ports were reset. Two vulnerabilities were detected:

- 80/tcp open http**: A Slowloris DOS attack vulnerability (CVE-2007-6750) was found. The state is 'LIKELY VULNERABLE'. The description states: 'Slowloris tries to keep many connections to the target web server open and hold them open as long as possible. It accomplishes this by opening connections to the target web server and sending a partial request. By doing so, it starves the http server's resources causing Denial Of Service.' The disclosure date is 2009-09-17. References include <http://ha.ckers.org/slowloris/> and <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750>.
- 443/tcp open https**: A Slowloris DOS attack vulnerability (CVE-2007-6750) was found. The state is 'LIKELY VULNERABLE'. The description states: 'Slowloris tries to keep many connections to the target web server open and hold them open as long as possible. It accomplishes this by opening connections to the target web server and sending a partial request. By doing so, it starves the http server's resources causing Denial Of Service.'

Other scripts like `http-csrf`, `http-stored-xss`, `http-slowloris-check`, `http-dombased-xss`, and `http-vuln-cve2014-3704` did not find any vulnerabilities or failed to execute.

```

|      Disclosure date: 2009-09-17
|      References:
|      http://ha.ckers.org/slowloris/
|      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
515/tcp open  printer
631/tcp open  ipp
|_http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
9100/tcp open  jetdirect
9877/tcp open  x510
MAC Address: 6C:F2:D8:04:7D:D5 (Canon)

Nmap done: 1 IP address (1 host up) scanned in 365.07 seconds

```

Étape 4 : Analyse

Lorsque nous analysons le resultat du scan nmap --script vuln 192.168.2.14 nous avons

➤ Ports ouverts

Port	État	Service
80/tcp	ouvert	HTTP
443/tcp	ouvert	HTTPS
515/tcp	ouvert	Printer (LPD)
631/tcp	ouvert	IPP
9100/tcp	ouvert	JetDirect
9877/tcp	ouvert	x510 (Canon propriétaire)

➤ Services actifs identifiés

- HTTP/HTTPS (80, 443) interface web d'administration de l'imprimante
- LPD (515) protocole d'impression classique
- IPP (631) Internet Printing Protocol, standard moderne
- JetDirect (9100) impression directe en réseau
- x510 (9877) service propriétaire Canon

➤ Vulnérabilité détectée

- Slowloris DOS CVE-2007-6750
- Statut : LIKELY VULNERABLE sur les ports 80 et 443

Reponse aux questions

1. Le nombre d'hotes détectés est 4
2. Sur l'imprimante Canon notre machine cible (192.168.2.14), 6 ports sont ouverts : 80, 443, 515, 631, 9100 et 9877.
3. Imprimante : HTTP, HTTPS, LPD, IPP, JetDirect et un service propriétaire Canon.
4. Oui, une vulnérabilité a été identifiée sur l'imprimante : Slowloris DoS (CVE-2007-6750), présente sur les ports 80 et 443
5. Le scan Nmap est un outil fondamental de reconnaissance et d'audit réseau. Il permet d'inventorier tous les appareils actifs, d'identifier les services exposés et de détecter les vulnérabilités avant qu'un attaquant ne le fasse. Dans une stratégie de cybersécurité, il s'inscrit dans la phase de tests de pénétration (pentest) et aide à réduire la surface d'attaque en identifiant les ports inutiles à fermer et les mettront à jour.